

Exercises 09/14

1. Let $f : (0, \infty) \rightarrow \mathbb{R}$ be defined by $f(x) = \frac{1}{x}$. Show that $f'(x) = -\frac{1}{x^2}$.
2. Show that if $f : \mathbb{R} \rightarrow \mathbb{R}$ is differentiable and $f' > 0$ everywhere then f is invertible.
3. What is the geometric interpretation of f'' ?
4. Suppose $f, g : [a, b] \rightarrow \mathbb{R}$. Show that if $f' = g'$ then there is a constant α such that $f(x) = g(x) + \alpha$.
5. Suppose that you know the change in the price level at time t is given by βt , where $\beta > 0$. What is the price level in time $t = 10$? Here you should assume that the price level is captured by a differential function $p : \mathbb{R}_+ \rightarrow \mathbb{R}_+$.
6. Show that $\frac{d \log(x)}{dx} = \frac{1}{x}$.
7. Show that if $f : \mathbb{R} \rightarrow \mathbb{R}$ is such that $f'(x) = f(x)$ then $f(x) = \alpha \exp(x)$ for some $\alpha \in \mathbb{R}$.
8. Let $f(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!}$. Here $0! \equiv 1$. Assume this sum converges for all x and that you are free to differentiate across an infinite sum. Compute $f'(x)$.
9. Show the following:
 - a. $x^\alpha x^\beta = x^{\alpha+\beta}$
 - b. $x^{-\alpha} = \frac{1}{x^\alpha}$
 - c. $\frac{dx^\alpha}{dx} = \alpha x^{\alpha-1}$
 - d. $(x^\alpha)^\beta = x^{\alpha\beta}$
 - e. $\log(x^r) = r \log(x)$
10. If $f : \mathbb{R} \rightarrow \mathbb{R}$ and $y = f(x)$ define the elasticity of y with respect to x as $\varepsilon_{yx} = \frac{dy}{dx} \frac{x}{y}$. Intuitively the elasticity is the percent change in y given a percent change in x ; the advantage elasticity is that it's a unit-free measure of change. Let $f(x) = x^\alpha$ for some $\alpha \in \mathbb{R}$.
 - a. Compute ε_{yx} .
 - b. Let $\hat{y} = \log y$ and $\hat{x} = \log x$. Compute $\frac{d\hat{y}}{d\hat{x}}$.